IAN MCPHERSON

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EDUCATION

Johns Hopkins University *August 2022 - Present*
Ph.D. in Applied Mathematics and Statistics
Current Advisor: Mauro Maggioni
Overall GPA: 4.00

Tufts University *August 2020 - May 2022*
M.Sc. in Mathematics
Concentration in Probability and Computational Methods
Overall GPA: 3.99

Occidental College *August 2015 - May 2019*
B.A. in Biochemistry & Economics

RESEARCH INTERESTS

High-Dimensional Statistics, Manifold Learning, Riemannian Optimization, Scaling Limits for Neural Networks

TEACHING

Johns Hopkins University

Fall 2024	**Instructor of Record** EN.500.111 Geometric Toolbox for Data Science Designed and conducted a survey course in Geometric Methods for High Dimensional Statistics.
Fall 2024	**Teaching Assistant** EN.553.634 Elements of Statistical Learning Undergraduate/graduate level course on the theory behind statistical learning
Summer 2024	**Instructor** Master's Program Probability Review Designed and gave a review series on probability fundamentals for incoming Master's students
Spring 2024	**Teaching Assistant** 553.738 for Professor *Mauro Maggioni* Graduate-level course on High-Dimensional Approximation, Probability, and Statistical Learning
Fall 2023	**Teaching Assistant** 553.632 for Professor *Sergey Kushnarev* Undergraduate/graduate-level course on Bayesian statistics
Summer 2023	**Instructor** Master's Program Probability Review Designed and gave a review series on probability fundamentals for incoming Master's students
Spring 2023	**Teaching Assistant** 553.681 Numerical Analysis for Professor *Mario Micheli* Undergraduate/graduate-level course on numerical analysis
Fall 2022	**Teaching Assistant** 553.691 Dynamical Systems for Professor *Yannis Kevrekedis* Undergraduate/graduate-level course on nonlinear dynamical systems

Tufts University

Summer 2022	**Teaching Assistant and Guest Lecturer** Math 70 Linear Algebra for *Curtis Heberle* Undergraduate introduction to linear algebra, giving five guest lectures
Spring 2022	**Teaching Assistant** Math 166 Statistics for Professor *Bruce Boghosian* Undergraduate mathematical statistics introduction
Fall 2021	**Teaching Assistant** Math 126 Numerical Linear Algebra for Professor *Abiy Tassisa* Undergraduate introduction to numerical linear algebra
Summer 2021	**Teaching Assistant and Guest Lecturer** Math 21 Introductory Statistics for *Linda Garant* Undergraduate introduction to statistics, giving four guest lectures

AWARDS AND SCHOLARSHIPS

Whiting School of Engineering Teaching Assistant Award *April 2024*
Sole Recipient across WSE for 2024, in recognition of excellence and talent as a graduate TA.
Rufus P. Isaacs Graduate Fellowship *Fall 2023-Spring 2024*

CONFERENCES AND WORKSHOPS

August 2024	Funded for Princeton Machine Learning Theory Summer Graduate School
June 2024	Funded for SLMath (UC Berekley) Summer Graduate School
	Interacting Particle Systems: Analysis and Computation
May 2024	Funded ICERM Workshop Participant
	Interacting Particle Systems: Analysis, Control, Learning and Computation
June 2023	Funded ICERM Workshop Participant
	Modern Applied and Computational Analysis
May 2023	Funded ICERM Workshop Participant: Optimal Transport
January 2023	Joint Mathematical Meetings Annual Conference

TALKS

January 2024	"Riemannian Optimization and Wasserstein Barycenters",
	Applied Mathematics and Statistics Student Seminar, JHU

PROFESSIONAL SERVICE AND COMMUNITY INVOLVEMENT

Fall 2023 - Present	Graduate Student Seminar Co-organizer
Fall 2023 - Present	Foundations of Data Science Journal Referee (1)
Fall 2023	Ph. D. Committee Member for an External Review of JHU's AMS Department

EXPERIENCE

MIT Lincoln Laboratories - Group 33 May 2022 - August 2022
Research Intern *Lexington, MA*

- Obtained DoD Security Clearance: Secret Level
- Developed algorithms for the classification of RF signals in Python.
- Verified the well-posedness of a MILP Battle Management Optimization problem, and compared and contrasted different MILP solvers within MATLAB.
- Field work in White Sands Missle Range for testing different radar array parts, as well as assisting with the teardown and packaging of a few sites.

Tufts University's Data Intensive Studies Center January 2022 - May 2022
DISC Research Intern *Medford, MA*

- Research assistant to Dr. Abani Patra, working on understanding and implementation of Markov Random Fields with an eye towards analysis of polar caps.

Tufts University's Math Department May 2021 - August 2021
Research Assistant *Medford, MA*

- Research assistant to Dr. Kasso Okoudjou, working with a set of orthogonal polynomials on the Sierpinski Gasket with an eye towards producing analogs to the Legendre Polynomials on the unit interval.

TECHNICAL QUALIFICATIONS

Proficient Coding Languages MATLAB, Python, Java, LaTeX.

HOBBIES

Dancing and choreographing, long distance running, rock climbing, and cooking.